Dataset Link : corona_virus_2020

Theoretical Questions

Question 1 : What is Data Loading in Power BI and why is it considered the first step of analysis?

Ans:-

Data Loading in Power BI refers to the process of importing data from various sources into Power BI so it can be used for analysis and reporting in Microsoft Power BI.

What is Data Loading?

It is the step where you:

- Connect to a data source (Excel, CSV, database, cloud, etc.)
- Import the data into Power BI Desktop
- Make the data available for transformation and visualization

In simple terms, it means bringing raw data into Power BI.

Why is it the First Step of Analysis?

1. Data is Required for Analysis

Without loading data, you cannot perform any analysis or create reports.

2. Foundation for All Next Steps

All later steps depend on this:

- Data cleaning (Power Query)
- Data modeling
- Creating visuals and dashboards

If data is not loaded, nothing else can be done.

3. Ensures Data Availability

It ensures that data from different sources is available in one place for analysis.

4. Defines Data Structure

When data is loaded, Power BI understands:

- Columns
- Data types
- Tables

This structure is necessary for accurate analysis.

Question 2 : Explain the difference between “Load” and “Transform Data” in Power BI.

Ans:-

1. Load

Load imports the data directly into Power BI without making any changes.

Purpose:

- Quickly bring data into Power BI
- Start creating reports and visualizations immediately

Example:

If your dataset is already clean and properly structured, you can simply click **Load**.

This option skips the data cleaning step.

2. Transform Data

Transform Data opens the Power Query Editor, where you can clean, modify, and prepare the data before loading it.

Common transformations:

- Remove duplicates
- Split or merge columns
- Change data types
- Filter rows
- Handle missing values

After making changes, you click Close & Apply to load the cleaned data.

Difference Between Load and Transform Data in Power BI

Feature	Load	Transform Data
Definition	Imports data directly into Power BI	Opens Power Query Editor to modify data
Purpose	Quick data loading	Data cleaning and transformation
Data Changes	No changes applied	Data can be cleaned and modified
Tool Used	Direct load into model	Uses Power Query Editor
When to Use	When data is already clean	When data needs preparation
Example	Load Excel file directly	Remove duplicates, split columns, etc.

Question 3 : What is a Fact Table and a Dimension Table? Give examples from the dataset.

Ans:-

Fact Table

A Fact Table contains quantitative (numeric) data that represents measurable events or metrics. It usually includes:

- Numbers you want to analyze (measures)
- Foreign keys linking to dimension tables

From your dataset:

The following columns are facts (measures) because they store numeric, analyzable data:

- Confirmed_Cases
- Recovered_Cases
- Deaths
- Active_Cases
- Tests_Conducted
- Hospitalized

Example Fact Record:

- Confirmed Cases = 10,000
- Deaths = 500
- Tests Conducted = 50,000

Dimension Table

A Dimension Table contains descriptive (textual or categorical) data that provides context to the facts. These are used for filtering, grouping, and labeling.

From your dataset:

The following columns act as dimensions:

- Country
- State
- Date
- Vaccination_Status

Example Dimension Record:

- Country = UK
- State = State_A

- Date = 2021-05-01
- Vaccination Status = Yes

Question 4 : Why is Star Schema preferred over Snowflake Schema in Power BI?

Ans:-

In tools like Power BI, the Star Schema is generally preferred over the Snowflake Schema because it is simpler, faster, and more efficient for analysis.

1. Better Performance

- In a **Star Schema**, dimension tables are denormalized (not split).
- This reduces the number of joins needed during queries.

Result:

- Faster data retrieval
- Better report performance in Power BI

2. Simpler Data Model

- Star Schema has:
    - One central Fact Table
    - Directly connected Dimension Tables
- Snowflake Schema has multiple related dimension tables (more complex).

Result:

- Easier to understand and design
- Less confusion when building reports

3. Easier for Users (Self-Service BI)

- Power BI is designed for business users, not just developers.
- Star Schema is intuitive:
    - Clear relationships
    - Simple drag-and-drop fields

Result:

- Non-technical users can create dashboards easily

4. Better DAX Performance

- DAX calculations work more efficiently with fewer relationships.

- Snowflake Schema introduces:
    - More joins
    - Complex filtering paths

Result:

- Faster and simpler DAX formulas

5. Optimized for Columnar Storage

- Power BI uses a columnar engine (VertiPaq).
- Denormalized tables (Star Schema) compress better.

Result:

- Reduced memory usage
- Faster processing

6. Fewer Relationship Issues

- Snowflake Schema may create:
    - Ambiguous relationships
    - Complex filter directions

Result:

- Star Schema avoids these issues with clean, single-level relationships

Practical Questions

Question 5 : Identify and remove duplicate records based on Date, Country, and State.

Ans:-

Goal

Remove duplicate rows based on:

- **Date**
- **Country**
- **State**

Step-by-Step in Power BI

Step 1: Open Power BI & Load Data

1. Open Power BI Desktop
2. Click Home → Get Data
3. Choose Text/CSV
4. Select your file → Click Load

Step 2: Open Power Query Editor

1. Go to Home tab
2. Click Transform Data

This opens Power Query Editor

Step 3: Select Required Columns

Now you'll choose the columns that define duplicates.

1. Hold CTRL
2. Click these columns:
   - **Date**
   - **Country**
   - **State**

Step 4: Remove Duplicates

With those columns selected:

1. Go to Home tab
2. Click Remove Rows
3. Select Remove Duplicates

What this does:

- Keeps the first occurrence
- Removes all repeated combinations of Date + Country + State

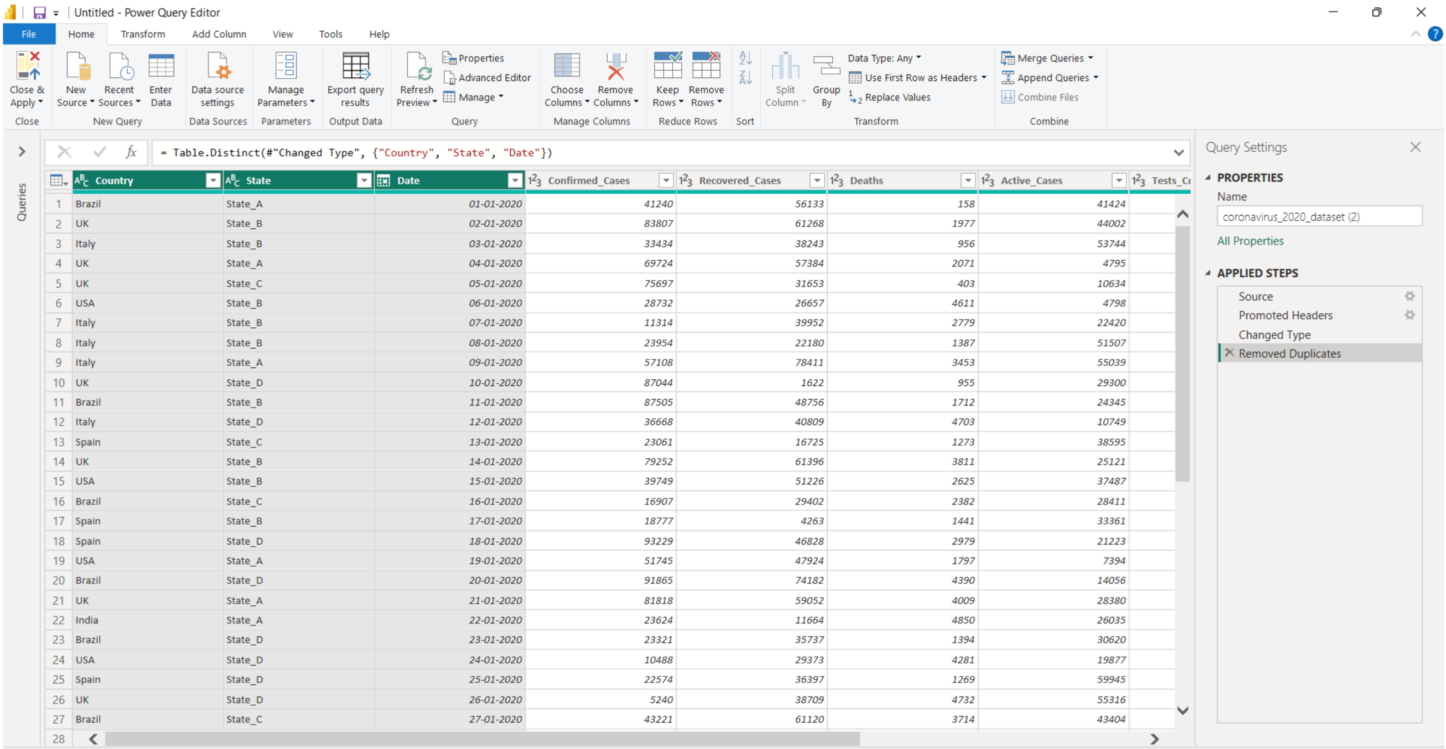
Step 5: Verify Result

- Check row count before and after
- Scroll and confirm duplicates are gone

Step 6: Apply Changes

1. Click Close & Apply
2. Power BI will update your dataset

Final Ans:-



	Country	State	Date	Confirmed_Cases	Recovered_Cases	Deaths	Active_Cases	Tests_C
1	Brazil	State_A	01-01-2020	41240	56133	158	41424	
2	UK	State_B	02-01-2020	83807	61268	1977	44002	
3	Italy	State_B	03-01-2020	33434	38243	956	53744	
4	UK	State_A	04-01-2020	69724	57384	2071	4795	
5	UK	State_C	05-01-2020	75697	31653	403	10634	
6	USA	State_B	06-01-2020	28732	26657	4611	4798	
7	Italy	State_B	07-01-2020	11314	39952	2779	22420	
8	Italy	State_B	08-01-2020	23954	22180	1387	51507	
9	Italy	State_A	09-01-2020	57108	78411	3453	55039	
10	UK	State_D	10-01-2020	87044	1622	955	29300	
11	Brazil	State_B	11-01-2020	87505	48756	1712	24345	
12	Italy	State_D	12-01-2020	36668	40809	4703	10749	
13	Spain	State_C	13-01-2020	23061	16725	1273	38595	
14	UK	State_B	14-01-2020	79252	61396	3811	25121	
15	USA	State_B	15-01-2020	39749	51226	2625	37487	
16	Brazil	State_C	16-01-2020	16907	29402	2382	28411	
17	Spain	State_B	17-01-2020	18777	4263	1441	33361	
18	Spain	State_D	18-01-2020	93229	46828	2979	21223	
19	USA	State_A	19-01-2020	51745	47924	1797	7394	
20	Brazil	State_D	20-01-2020	91865	74182	4390	14056	
21	UK	State_A	21-01-2020	81818	59052	4009	28380	
22	India	State_A	22-01-2020	23624	11664	4850	26035	
23	Brazil	State_D	23-01-2020	23321	35737	1394	30620	
24	USA	State_D	24-01-2020	10488	29373	4281	19877	
25	Spain	State_D	25-01-2020	22574	36397	1269	59945	
26	UK	State_D	26-01-2020	5240	38709	4732	55316	
27	Brazil	State_C	27-01-2020	43221	61120	3714	43404	
28								

Question 6 : Identify and replace null values in Vaccination_Status.

Ans:-

Goal

Identify and replace null values in Vaccination_Status

STEP 1: Load your data

1. Open Power BI Desktop
2. Click Home → Get Data → Text/CSV
3. Select your CSV file
4. Click Load

STEP 2: Open Power Query Editor

1. Go to Home tab
2. Click Transform Data

This opens the Power Query Editor, where all cleaning happens.

STEP 3: Locate the column

1. In the table preview, find the column:

Vaccination_Status

STEP 4: Identify null values

There are 2 easy ways:

Method A (Filter method)

1. Click the **dropdown arrow** on Vaccination_Status
2. Look for (null) in the list
3. If it appears → you have null values

Method B (Column profiling – better)

1. Go to View tab
2. Enable:
    - Column quality
    - Column distribution

You'll see:

- Valid %
- Error %
- Empty % (this = null values)

STEP 5: Replace null values

Now we fix them

Option A: Replace with a value (most common)

1. Select the column Vaccination_Status
2. Go to Transform tab
3. Click Replace Values
4. In the popup:
    - **Value to Find** → type:

      null

    - **Replace With** → type something like:
        - Not Vaccinated
        - Unknown
        - No Data

5. Click OK

Important Tip:

If typing null doesn't work:

Use this instead:

1. Click dropdown on column
2. Uncheck everything except (null)
3. Right-click the column → Replace Values
4. Replace with your desired value

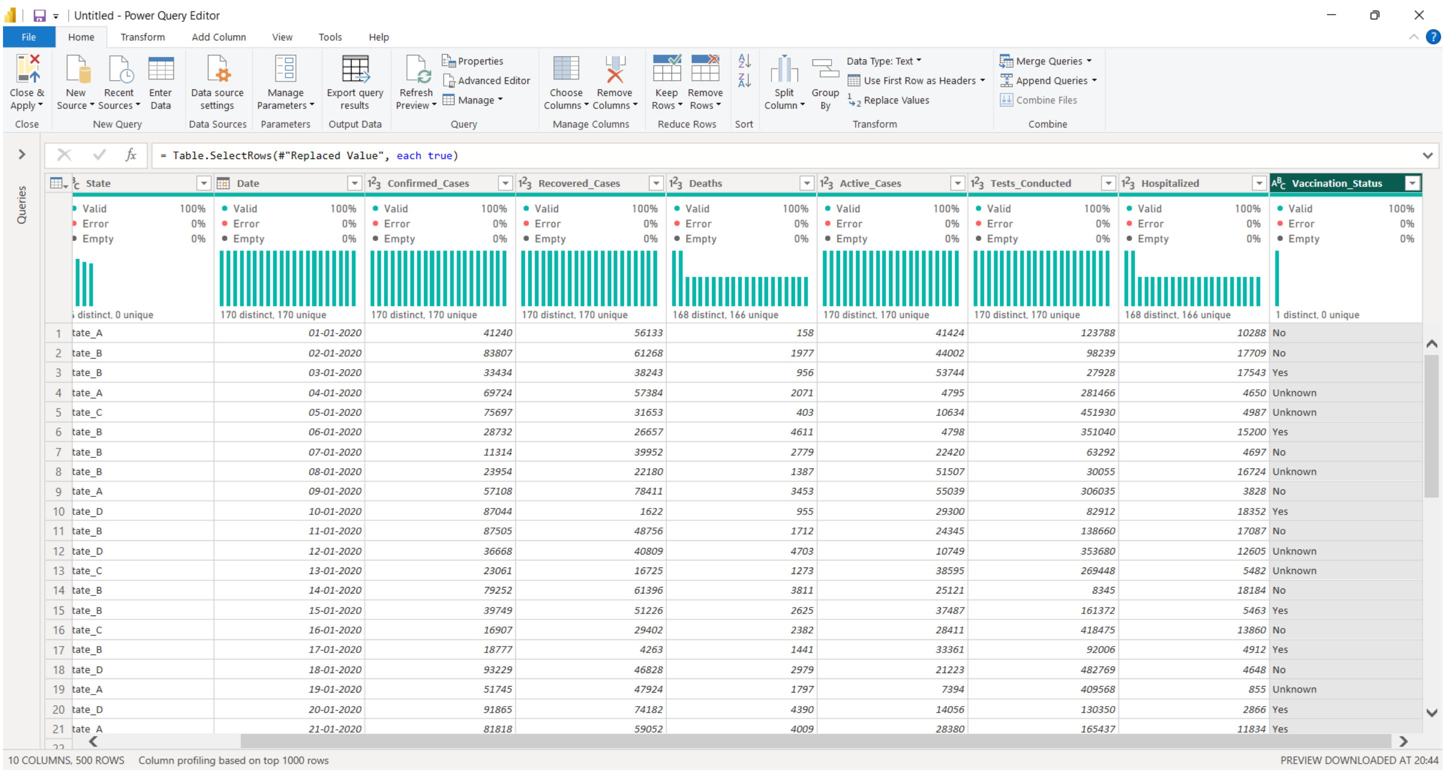
STEP 6: Verify changes

1. Check the column again
2. Ensure:
    - No (null) values remain
    - Your replacement text appears

STEP 7: Apply changes

1. Click Home → Close & Apply

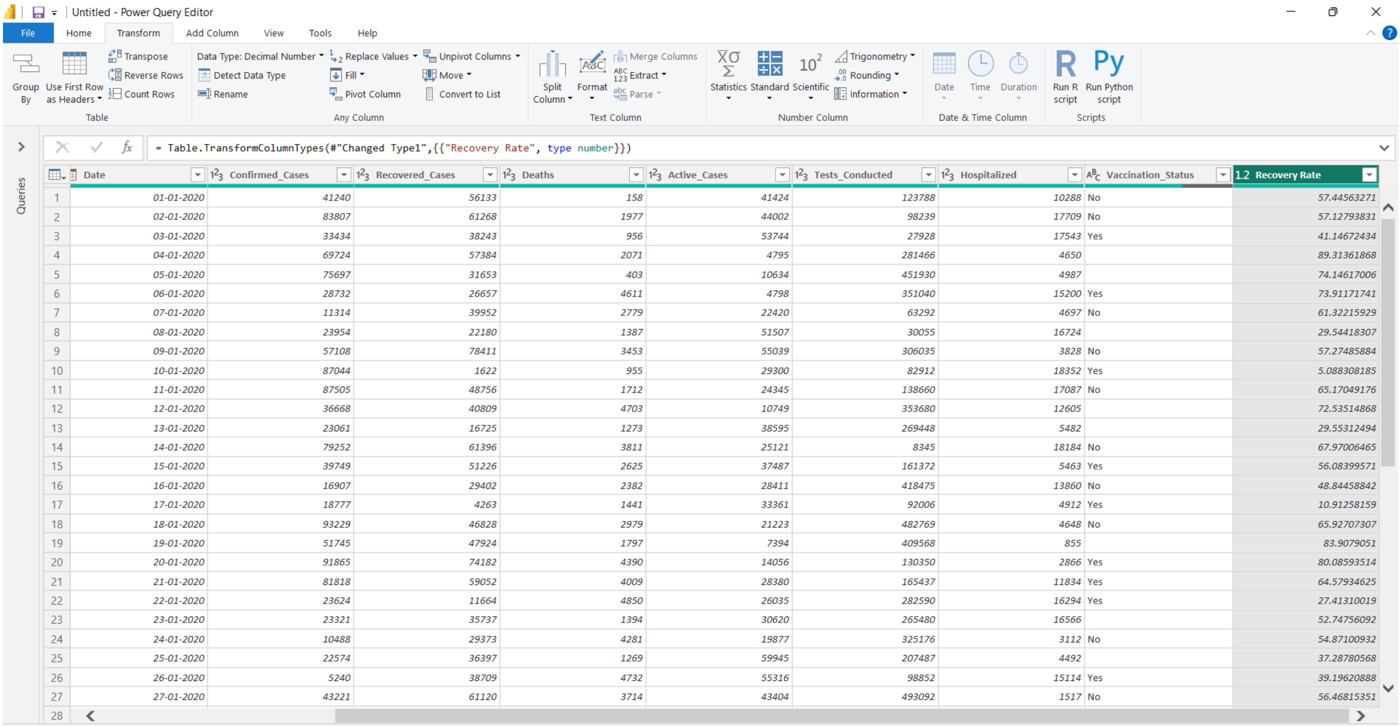
Final Ans:-



	State	Date	Confirmed_Cases	Recovered_Cases	Deaths	Active_Cases	Tests_Conducted	Hospitalized	Vaccination_Status
1	tate_A	01-01-2020	41240	56133	158	41424	123788	10288	No
2	tate_B	02-01-2020	83807	61268	1977	44002	98239	17709	No
3	tate_B	03-01-2020	33434	38243	956	53744	27928	17543	Yes
4	tate_A	04-01-2020	69724	57384	2071	4795	281466	4650	Unknown
5	tate_C	05-01-2020	75697	31653	403	10634	451930	4987	Unknown
6	tate_B	06-01-2020	28732	26657	4611	4798	351040	15200	Yes
7	tate_B	07-01-2020	11314	39952	2779	22420	63292	4697	No
8	tate_B	08-01-2020	23954	22180	1387	51507	30055	16724	Unknown
9	tate_A	09-01-2020	57108	78411	3453	55039	306035	3828	No
10	tate_D	10-01-2020	87044	1622	955	29300	82912	18352	Yes
11	tate_B	11-01-2020	87505	48756	1712	24345	138660	17087	No
12	tate_D	12-01-2020	36668	40809	4703	10749	353680	12605	Unknown
13	tate_C	13-01-2020	23061	16725	1273	38595	269448	5482	Unknown
14	tate_B	14-01-2020	79252	61396	3811	25121	8345	18184	No
15	tate_B	15-01-2020	39749	51226	2625	37487	161372	5463	Yes
16	tate_C	16-01-2020	16907	29402	2382	28411	418475	13860	No
17	tate_B	17-01-2020	18777	4263	1441	33361	92006	4912	Yes
18	tate_D	18-01-2020	93229	46828	2979	21223	482769	4648	No
19	tate_A	19-01-2020	51745	47924	1797	7394	409568	855	Unknown
20	tate_D	20-01-2020	91865	74182	4390	14056	130350	2866	Yes
21	tate_A	21-01-2020	81818	59052	4009	28380	165437	11834	Yes

Question 7 : Create a new column to calculate Recovery Rate.

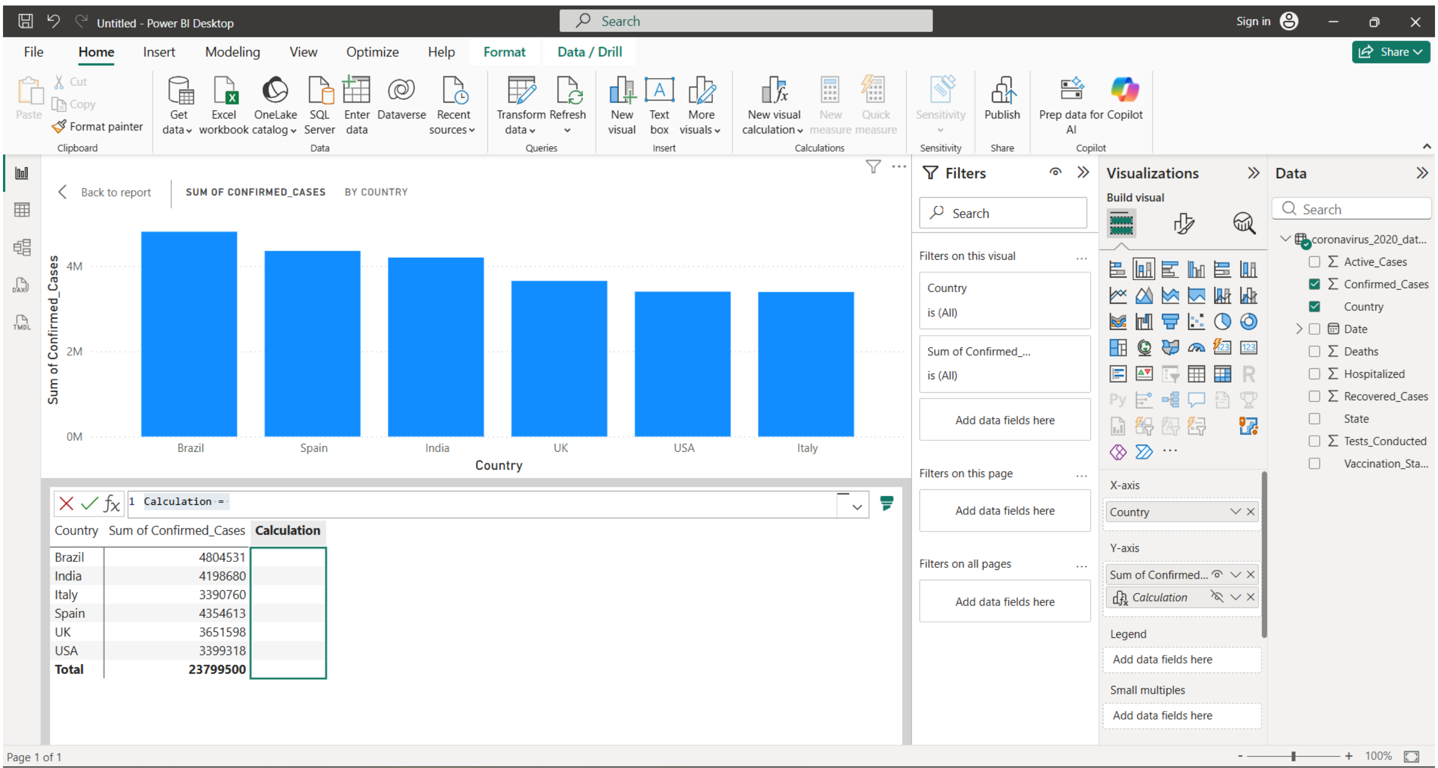
Ans:-



	Date	Confirmed_Cases	Recovered_Cases	Deaths	Active_Cases	Tests_Conducted	Hospitalized	Vaccination_Status	Recovery Rate
1	01-01-2020	41240	56133	158	41424	123788	10288	No	57.44563271
2	02-01-2020	83807	61268	1977	44002	98239	17709	No	57.12793831
3	03-01-2020	33434	38243	956	53744	27928	17543	Yes	41.14672434
4	04-01-2020	69724	57384	2071	4795	281466	4650		89.31361868
5	05-01-2020	75697	31653	403	10634	451930	4987		74.14617006
6	06-01-2020	28732	26657	4611	4798	351040	15200	Yes	73.91171741
7	07-01-2020	11314	39952	2779	22420	63292	4697	No	61.32215929
8	08-01-2020	23954	22180	1387	51507	30055	16724		29.54418307
9	09-01-2020	57108	78411	3453	55039	306035	3828	No	57.27485884
10	10-01-2020	87044	1622	955	29300	82912	18352	Yes	5.088308185
11	11-01-2020	87505	48756	1712	24345	138660	17087	No	65.17049176
12	12-01-2020	36668	40809	4703	10749	353680	12605		72.53514868
13	13-01-2020	23061	16725	1273	38595	269448	5482		29.55312494
14	14-01-2020	79252	61396	3811	25121	8345	18184	No	67.97006465
15	15-01-2020	39749	51226	2625	37487	161372	5463	Yes	56.08399571
16	16-01-2020	16907	29402	2382	28411	418475	13860	No	48.84458842
17	17-01-2020	18777	4263	1441	33361	92006	4912	Yes	10.91258159
18	18-01-2020	93229	46828	2979	21223	482769	4648	No	65.92707307
19	19-01-2020	51745	47924	1797	7394	409568	855		83.9079051
20	20-01-2020	91865	74182	4390	14056	130350	2866	Yes	80.08593514
21	21-01-2020	81818	59052	4009	28380	165437	11834	Yes	64.57934625
22	22-01-2020	23624	11664	4850	26035	282590	16294	Yes	27.41310019
23	23-01-2020	23321	35737	1394	30620	265480	16566		52.74756092
24	24-01-2020	10488	29373	4281	19877	325176	3112	No	54.87100932
25	25-01-2020	22574	36397	1269	59945	207487	4492		37.28780568
26	26-01-2020	5240	38709	4732	55316	98852	15114	Yes	39.19620888
27	27-01-2020	43221	61120	3714	43404	493092	1517	No	56.46815351

Question 8 : Create a summarized table showing total confirmed cases by Country.

Ans:-



Country	Sum of Confirmed_Cases	Calculation
Brazil	4804531	
India	4198680	
Italy	3390760	
Spain	4354613	
UK	3651598	
USA	3399318	
Total	**23799500**	